

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9648/02

Paper 2 Core Paper

1 September 2014

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

Circle the question number of the question attempted

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
8 / 9	
Total	

This document consists of printed **23** pages.

[Turn over

Section A

Answer **all** the questions in this section.

- 1 (a) Fig.1.1 shows the relationship between certain organelles in a phagocytic cell as it engulfs particles into the cell.

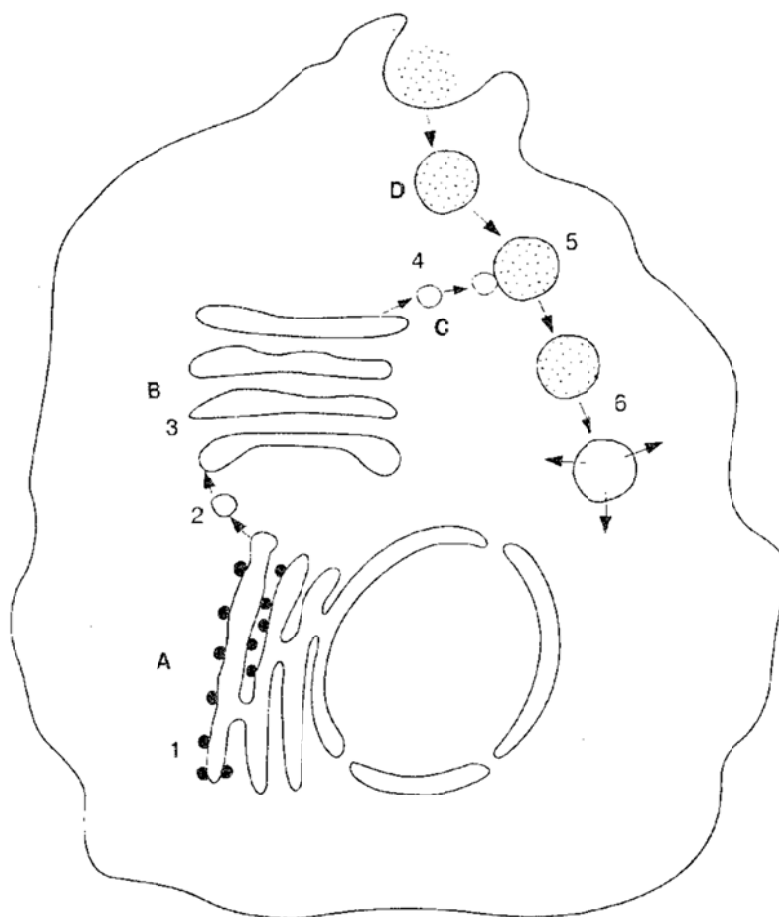


Fig. 1.1

- (i) Identify organelles **A** to **D**. [2]
- A** rough endoplasmic reticulum
- B** Golgi apparatus/Golgi body
- C** lysosome
- D** endocytic vesicle/food vacuole

- (ii) Number 1 to 6 on Fig. 1.1 represents a sequence of events occurring in the cell. State what is occurring at 1 to 6. [6]

- 1 **synthesis of proteins at ribosomes, products passed into cisternae/lumen of rough ER where they are modified by enzymes/glycosylation;;**
- 2 **transport vesicle containing synthesized proteins pinched off from rough ER and fuses with cisternae at *cis* face of Golgi apparatus;;**
- 3 **protein undergoes further modification, sorting and packaging (e.g. glycosylation, phosphorylation) to form glycoproteins etc;;**
- 4 **lysosome buds off the cisternae at *trans* face of Golgi apparatus;;**
- 5 **lysosome fuses with endocytic vesicle and contents of endocytic vesicle undergo intracellular digestion by hydrolytic enzymes / proteases;;**
- 6 **useful products of digestion reabsorbed/assimilated;;**

(b) Fig. 1.2 shows a diagrammatic representation of a section of the nucleus in a cell.

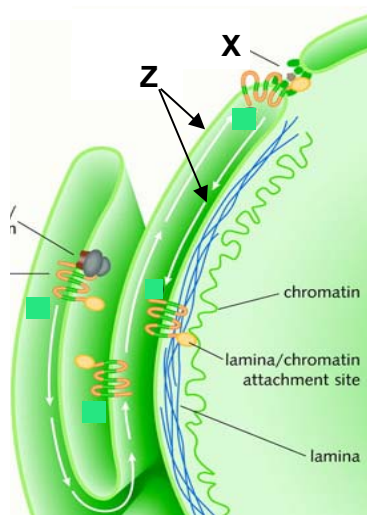


Fig. 1.2

- (i) Briefly describe the role of X. [1]
It allows the exit of mRNA from the nucleus/the entry of nucleotides into the nucleus;;
- (ii) Identify Z. [1]
Nuclear envelope;;
- (iii) Briefly describe the structure of Z. [2]
1. Nuclear envelope made up of two phospholipid bilayers;;
2. where they are arranged such that the hydrophilic phosphate heads are facing the external and internal aqueous environment while hydrophobic hydrocarbon tails are kept within the bilayer;;

[Total:12]

- 2 Fig. 2.1 shows a section of a DNA double helix, comprising of 2 complementary polynucleotide chains wound around each other.

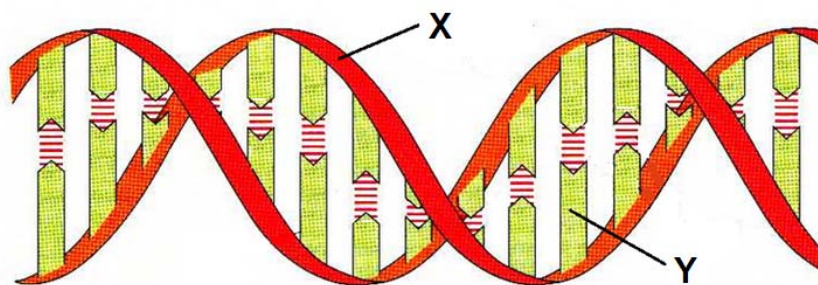


Fig. 2.1

- (a) (i) With reference to Fig. 2.1, label the components **X** and **Y**. [1]

X - sugar-phosphate backbone;

Y - nitrogenous base;

- (ii) State one way in which the structure of DNA differs from RNA. [1]

1. **DNA is double-stranded while RNA is single-stranded;;**
2. **The pentose sugar in DNA nucleotide is a deoxyribose while the pentose sugar in RNA nucleotide is a ribose;;**
3. **The nitrogenous bases in DNA can only be adenine, cytosine, guanine or thymine while the nitrogenous bases in RNA can only be adenine, cytosine, guanine or uracil;;**

[any one]

DNA contains the genetic information needed for protein synthesis. Fig. 2.2 shows a stage during translation.

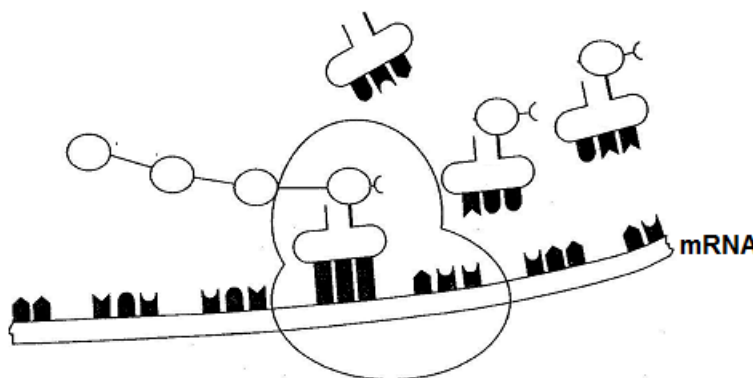


Fig. 2.2

- (b) Explain how tRNA binds to mRNA. [2]

1. **complementary base-pairing via hydrogen bond formation;;**
2. **between codon on mRNA and anticodon on tRNA;;**
3. **adenine-uracil and guanine-cytosine;;**

(c) Describe how the process shown in Fig. 2.2 is terminated. [4]

1. **Elongation continues until a stop codon, (UAA/ UGA/ UAG) on mRNA reaches A site of the ribosome;;**
2. **Protein release factor binds to stop codon;;**
3. **causing addition of a water molecule to polypeptide chain which hydrolyses the completed polypeptide from the tRNA in P site;;**
4. **freeing the polypeptide from the ribosome and the remainder of translational complex are disassembled;;**

(d) Explain why a single base-pair deletion in the DNA sequence is more likely to produce a non-functional protein compared to a single base-pair substitution. [3]

1. **Base-pair deletion results in frameshift mutation where the reading frame is altered / codons downstream of the mutation are read in different set of three's;;**
2. **results in different amino acids coded for with different properties due to different R groups / change in amino acid sequence resulting in changes to 3D conformation of protein and loss of protein function;;**
3. **Base-pair substitution usually result in silent mutation resulting in the same amino acid being coded for due to the degeneracy of the genetic code;;**
OR
4. **Missense mutation resulting in changes to only a single amino acid;;**

[Total: 11]

3 Fig. 3.1 shows a Human Immunodeficiency Virus (HIV) infecting a T cell.

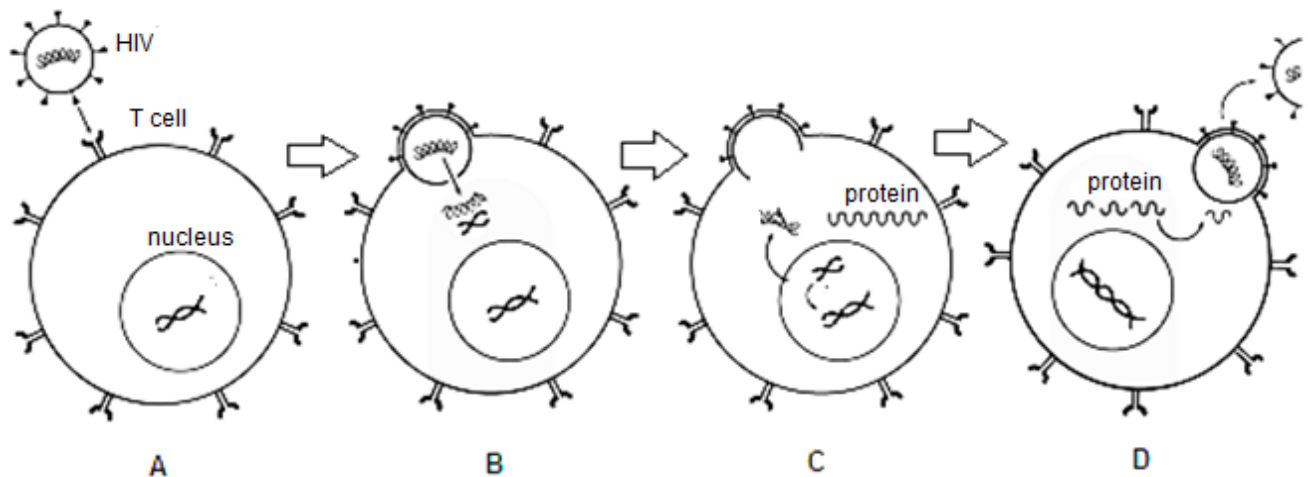


Fig. 3.1

(a) Describe how HIV enters T cell. [3]

1. gp120 on HIV envelope binds to a CD4 on the plasma membrane of the T cell;
2. interaction between gp120 and CD4 induces a conformation change in gp120, enabling the binding of gp120 to T cell's CCR5/CXCR4;
3. interaction between gp120 and chemokine receptor brings about a conformation change in gp41 on the HIV virion, allowing the fusion of the HIV envelope and T cell membrane;;

(b) Outline the role(s) of a named enzyme in:

(i) stage **B** of the HIV life cycle. [2]

1. reverse transcriptase (RT) catalyses the synthesis of a cDNA strand complementary to viral RNA;;
2. RT catalyses the synthesis of a complementary DNA strand to cDNA forming a double stranded (ds) DNA;;

(ii) stage **C** of the HIV life cycle [1]

3. dsDNA is incorporated into the host cell's DNA as a provirus by integrase;;

Fig. 3.2 shows the electron micrograph of stage **D**.

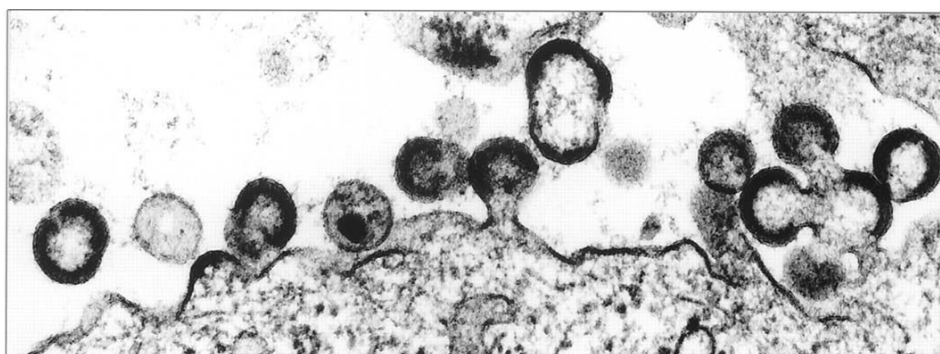


Fig. 3.2

Saquinavir, an anti-HIV drug is known to inhibit enzymatic activity in the maturation and release of HIV virions as seen in stage **D**.

(c) Identify the enzyme inhibited and explain how the use of Saquinavir leads to HIV infection being controlled. [3]

1. **Saquinavir binds to the active site of HIV protease;;**
2. **HIV polypotein cannot be cleaved into functional protein- capsid proteins, reverse transcriptase and integrase (any 2);;**
3. **mature virion cannot be formed and infection is prevented from spreading;;**

HIV infection is usually life-threatening.

(d) Explain how HIV infection causes death in humans. [2]

1. **When HIV infects the T helper cells, replication of the virus destroys T cells, resulting in low T cell count;;**
2. **When the T cell count gets very low in an affected person, the immune system becomes so weak that one becomes susceptible to opportunistic infections which can cause death in humans;;**

[Total: 11]

- 4 Breast cancer 1, (*BRCA 1*) gene is a human tumour suppressor gene. Its protein product, breast cancer type 1 susceptibility protein (*BRCA 1*) is responsible for repairing DNA damage.

Fig. 4 shows the arrangement of *BRCA1* gene with respect to its core promoter (black horizontal block). The numbers indicate the nucleotides upstream or downstream of the transcription start site (indicated with an arrow).

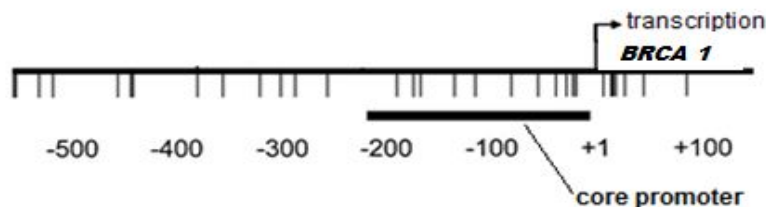


Fig.4

In normal human mammary epithelial cells, the core promoter region was found to be unmethylated, however, unusual cytosine methylation of this region was found in sporadic (non-hereditary) cases of breast cancer.

- (a) State what is meant by a tumour suppressor gene. [1]

1. **codes for proteins that inhibit cell cycle;;**

- (b) Describe the role of the core promoter in the expression of *BRCA 1* gene. [3]

1. **TATA-binding protein / TF IID recognizes and binds to TATA box within the core promoter region;;**
2. **General transcription factors bind to the TATA-binding protein recruiting RNA polymerase;;**
3. **RNA polymerase recognizes and binds to the promoter and assemble to form the transcription initiation complex initiating transcription;;**

- (c) Explain the effect that methylation of the core promoter of *BRCA 1* has on cancer development. [4]

1. **Methylation of promoter sequence makes the chromatin structure more compact;;**
2. **Promoter is inaccessible to RNA polymerase / GTFs / transcription initiation complex cannot assemble;;**
3. **hence *BRCA 1* gene is not expressed/ transcribed/ transcribed and translated;;**
4. ***BRCA 1* protein which is responsible for repairing DNA damage is not produced and mutation/damage goes unrepaired/unchecked;;**
5. **This gives rise to an accumulation of cancer-causing mutations;;**

Hereditary breast cancer, unlike sporadic breast cancer, is often associated with *BRCA 1* gene mutation in the germline.

- (d) State and explain how one named factor that can give rise to *BRCA1* gene mutation in the germline. [2]

1. **(Exposure to) radiation e.g. X-ray, ionising radiation / carcinogens e.g. tar / viruses;; Specific carcinogen required**
2. **Causes base-pair substitution/insertion/deletion/inversion;;**

(e) Suggest how *BRCA 1* gene product formation in sporadic and hereditary breast cancer differs. [1]

1. **No BRCA 1 protein is formed in sporadic breast cancer while in hereditary breast cancer, BRCA 1 protein formed (may be truncated/non-functional);;**

[Total: 11]

- 5 A team of geneticists was examining the inheritance of flower colour and vine height in pea plants. Pure-breeding pea plants with blue flowers and tall vines were crossed with pure-breeding pea plants with white flowers and short vines. All the pea plants in the F1 generation had blue flowers and tall vines. The F1 plants were selfed to give the F2 generation.

- (a) State the expected F2 ratio, assuming that the two genes are on separate chromosomes.

[1]

9:3:3:1

The actual results of the F2 generation were as follows:

Blue flower, tall vine	136
Blue flower, short vine	70
White flower, tall vine	62
White flower, short vine	4

- (b) Using the symbols **B** and **b** for flower colour and **T** and **t** for vine height, draw a genetic diagram to explain the observed results of the F2 generation. [5]

Let **B** represent the dominant allele for blue flower.

Let **b** represent the recessive allele for white flower.

Let **T** represent the dominant allele for tall vine.

Let **t** represent the dominant allele for short vine.

} 1

F1 phenotypes:

**Blue flower,
tall vine**

X

**Blue flower,
tall vine**

} 1

F1 genotypes:

**BT
bt**

X

**BT
bt**

Gametes:

BT

Bt

BT

Bt

bt

bT

bt

bT

} 1

Punnett Square:

	$\textcircled{\underline{BT}}$	$\textcircled{\underline{bt}}$	$\textcircled{\underline{Bt}}$	$\textcircled{\underline{bT}}$
$\textcircled{\underline{BT}}$	$\frac{\underline{BT}}{\underline{BT}}$	$\frac{\underline{BT}}{\underline{bt}}$	$\frac{\underline{BT}}{\underline{Bt}}$	$\frac{\underline{BT}}{\underline{bT}}$
$\textcircled{\underline{bt}}$	$\frac{\underline{bt}}{\underline{BT}}$	$\frac{\underline{bt}}{\underline{bt}}$	$\frac{\underline{bt}}{\underline{Bt}}$	$\frac{\underline{bt}}{\underline{bT}}$
$\textcircled{\underline{Bt}}$	$\frac{\underline{Bt}}{\underline{BT}}$	$\frac{\underline{Bt}}{\underline{bt}}$	$\frac{\underline{Bt}}{\underline{Bt}}$	$\frac{\underline{Bt}}{\underline{bT}}$
$\textcircled{\underline{bT}}$	$\frac{\underline{bT}}{\underline{BT}}$	$\frac{\underline{bT}}{\underline{bt}}$	$\frac{\underline{bT}}{\underline{Bt}}$	$\frac{\underline{bT}}{\underline{bT}}$

1

F2 genotype :

$\frac{\underline{BT}}{\underline{BT}}$	:	$\frac{\underline{Bt}}{\underline{Bt}}$	:	$\frac{\underline{bT}}{\underline{bT}}$	:	$\frac{\underline{bt}}{\underline{bt}}$
$\frac{\underline{BT}}{\underline{Bt}}$		$\frac{\underline{Bt}}{\underline{bt}}$		$\frac{\underline{bT}}{\underline{bt}}$		
$\frac{\underline{BT}}{\underline{bT}}$						
$\frac{\underline{BT}}{\underline{bt}}$						

F2 phenotype :

Blue flower, tall vine	:	Blue flower, short vine	:	White flower, tall vine	:	White flower, short vine
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F2 phenotypic ratio :

34	:	18	:	16	:	1
136		70		62		4
68		35		31		2

- (c) A chi-squared test was carried out to find out if there is any significant difference between the observed and expected F2 phenotypic ratio.

The formula for χ^2 and the table of probabilities are given as follows:

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \nu = c - 1$$

where Σ = 'sum of...'

O = observed 'value'

E = expected 'value'

ν = degrees of freedom

c = number of classes

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(i) Complete the table below and calculate the χ^2 value. [1]

Phenotypes	Observed	Expected	(O-E)	(O-E) ² /E
Blue flower, tall vine	136	153	-17	1.889
Blue flower, short vine	70	51	19	7.078
White flower, tall vine	62	51	11	2.373
White flower, short vine	4	17	-13	9.941

1

$$\chi^2_{\text{calculated}} = 1.889 + 7.078 + 2.373 + 9.941 = 21.3 \text{ (3s.f.)}$$

(ii) At 5% level of significance, conclude the results of the chi-squared test. [4]

1. (DOF = 4-1 = 3), the calculated χ^2 value is **21.3**, more than the critical χ^2 value of **7.82** at p= 0.05;;
2. Value of p is less than 0.001, less than p = 0.05;;
3. There is significant difference between the observed and the expected ratio;;
4. Any deviation from the expected is not due to chance, but due to other factors such as linkage;;

(iii) Based on your answer in part (ii), explain the mode of inheritance for the two genes in these pea plants. [2]

1. The 2 genes for flower colour and vine height are linked/ found on the same chromosome;;
2. which means that they form a linkage group and are usually inherited together in the same gamete / do not undergo independent assortment and segregation during meiosis;; R : complete linkage
3. Allele for blue flower colour is dominant to that of white flower colour and allele for tall vine is dominant to that of short vine;;

[Total: 13]

- 6 ATP synthase is an enzyme that synthesizes ATP from ADP and inorganic phosphate (P_i). Fig. 6.1 shows the structure of ATP synthase. It is found in both the chloroplast and the mitochondrion.

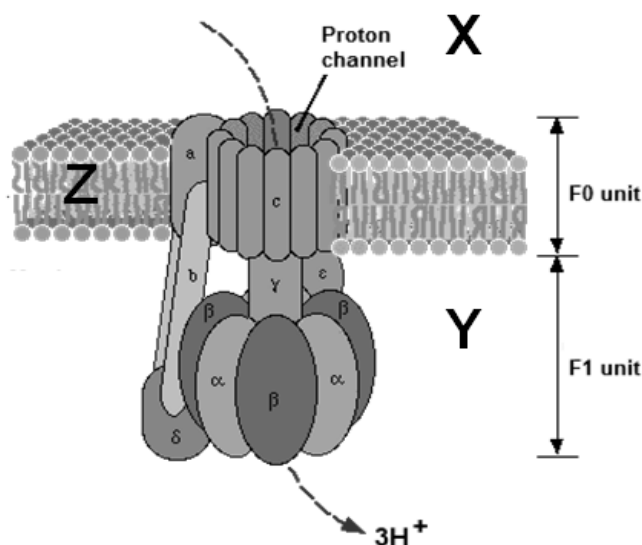


Fig. 6.1

- (a) Identify the regions (X, Y, Z) as found in the mitochondrion and chloroplast respectively.

	X	Y	Z
Mitochondrion	intermembrane space	matrix	inner mitochondrial membrane
Chloroplast	thylakoid space	stroma	thylakoid membrane

[2]

In both organelles, a proton gradient exists across the membrane where ATP synthase is located. The synthesis of ATP by ATP synthase is coupled with the diffusion of H^+ across the proton channel down its concentration gradient.

- (b) Describe how this proton gradient is generated in the chloroplast. [3]

1. Upon absorption of light, electron is displaced from a special chlorophyll a molecule / from the reaction centre of photosystems I and II;;
2. electron passes along a chain of electron carriers of the electron transport chain of progressively lower energy levels;;
3. energy released from electron transport is used to pump protons from stroma into thylakoid space, creating a proton gradient across the thylakoid membrane;;
4. Reference to thylakoid membrane being impermeable to H^+ ;;

The blue dye DCPIP can be converted to colourless reduced DCPIP by gaining electrons.

A suspension of chloroplasts was made by grinding fresh leaves in buffer solution and centrifuging the mixture. Tubes were then prepared and treated in the following way:

Tubes	Content	Treatment	Colour	
			Start	After 20 mins
A	1 cm ³ chloroplast suspension 5cm ³ DCPIP	Illuminated strongly	Blue-green	Green
B	1 cm ³ buffer solution 5cm ³ DCPIP	Illuminated strongly	Blue	Blue
C	1 cm ³ chloroplast suspension 5cm ³ DCPIP	Left in the dark	Blue-green	Blue-green

- (c) (i) With reference to tube **A**, name a possible source of electrons that reduces DCPIP. [1]

Any 1 of the answer: PSII/chlorophyll a, PSI, water

- (d) Explain this observation using your knowledge on light-dependent reactions of photosynthesis. [3]

- DCPIP acts like a final hydrogen ion and electron acceptor, like NADP⁺ in non-cyclic photophosphorylation of the light-dependent reaction;;**
- When light and chloroplast suspension are present, electrons from photolysis of water/ photo-excited electrons from PSII are passed through the electron transport chain in the thylakoid membranes and finally to NADP⁺ or DCPIP to reduce them;;**
- When the dye is reduced, colour of dye changes from blue to colourless;;**
- DCPIP is thus an indicator of whether light reaction and thus electron flow through the electron transport chain between PSII and PSI during light-dependent reactions has taken place or not;;**

- (e) Paraquat is a weedkiller. It binds with electrons of PS I.

Suggest how paraquat results in the death of a plant. [2]

- Prevents cyclic and non-cyclic photophosphorylation;;**
- No ATP and NADPH for Calvin cycle/conversion of GP to TP;;**
- No glucose made for respiration;;**

[Total: 11]

- 7 Fig. 7.1 shows the adult chorus cicada, *Amphipsalta zelandica*. This cicada is only found in New Zealand.



Fig. 7.1

- (a) Complete the table 7.2 to show the classification of the adult chorus cicada. [1]

Kingdom	Animalia
Phylum	Arthropoda
Class	Insecta
Order	Hemiptera
Family	Cicadidae
Genus	<u>Amphipsalta</u>
Species	<u>Amphipsalta zelandica</u>

Table 7.2

4 correct = 1 mark, 2-3 correct = 0.5 mark, 0-1 correct = 0 mark,

Evolutionary relationships between different species are investigated by examining the DNA sequences.

- (b) Explain how DNA sequences can be used to determine evolutionary relatedness between species. [2]
- 1. Compare regions of homology in the same gene (e.g. cytochrome c gene/ β -chain haemoglobin gene) found in different species/compare homologous genes between species;;**
 - 2. The fewer the differences in DNA sequences of homologous genes between species, the more closely related the species are/vice-versa;;**

Sections of DNA sequences in 14 species of cicada found in Australia, New Caledonia and New Zealand (1 to 14) were examined for similarities and differences.

The results of the DNA examination of these species were used to make a diagram showing the possible phylogenetic relationships of these cicadas as shown in Fig. 7.3 below.

Note that the cicada species in New Zealand are in two separate groups.

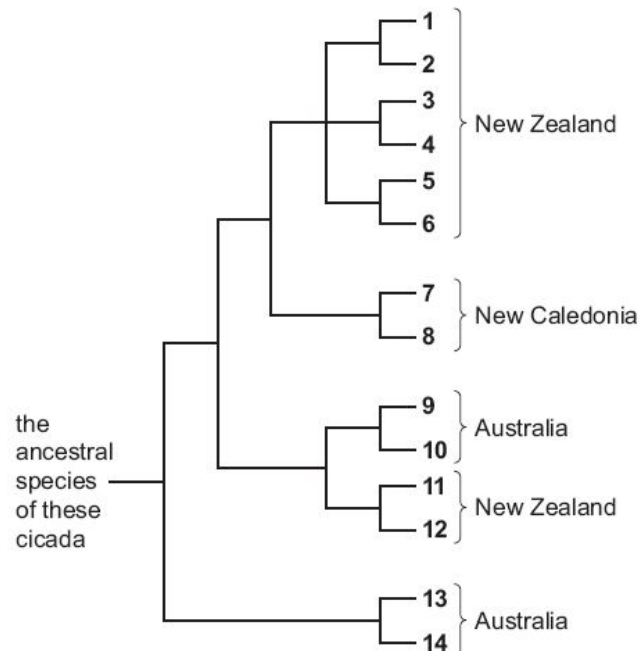


Fig. 7.3

It is suggested that the eight cicada species in New Zealand originated from two migrations, A and B, from Australia as shown in Fig.7.4.

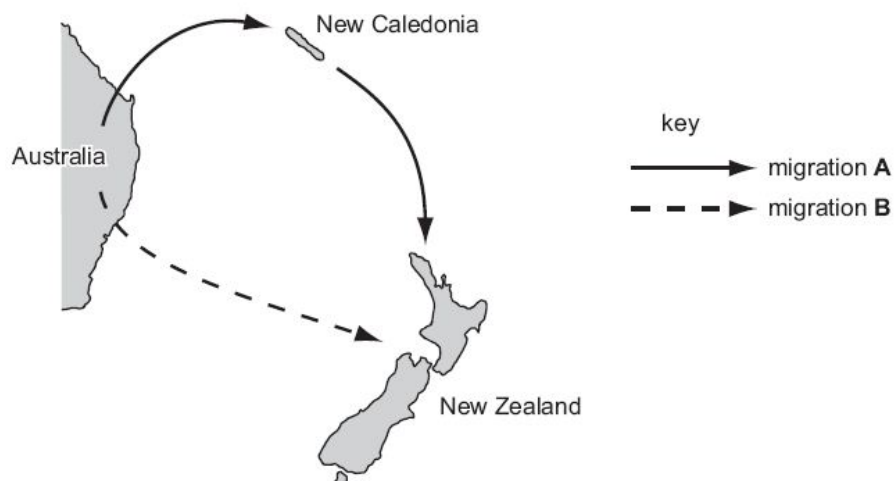


Fig. 7.4

(c) With reference to Fig. 7.3, explain how the data in the figure supports the theory that the eight species of cicada in New Zealand came about as a result of two migrations as shown in Fig. 7.4. [4]

1. **Species 1 – 6, New Caledonia / indirect / migration A;;**
2. **as shown by closer relatedness of the species 1-6 with the New Caledonian species 7-8;;**
3. **Species 11&12, direct (Australia) / migration B;;**
4. **as shown by closer relatedness with 9 &10 compared with 1-6;;**
5. **AVP making reference to Fig. 7.3;;**

Islands in the Pacific Ocean have been colonised by populations of animals that have migrated from Australia, mainland Asia and the Americas. Over many generations these populations have changed and become different species. Now they are unable to breed with animals of the original populations in Australia, mainland Asia and the Americas.

(d) Explain how natural selection has given rise to the changes in the populations of these animals on the Pacific islands. [4]

1. **Different environments on the different islands leads to different selection pressures on each of these islands;;**
2. **The islands were geographically isolated from the mainland and hence there was no interbreeding and gene flow was disrupted between the island sub-populations and the mainland populations;;**
3. **Variations present within the different island sub-populations, those with favourable characteristics were selected for;;**
4. **These individuals survived, reproduced and passed on the favourable genes to subsequent generations;;**
5. **Evolutionary changes happened independently in each sub-population, leading to different genetic changes in each sub-population;;**
6. **Over many thousands of successive generations, each sub-population becomes a different species / genetically distinct species;;**

Points 1, 3, 4 and 5 required for full credit

[Total:12]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8 (a) Explain, with examples, the structural differences between fibrous and globular proteins.
[6]

Named examples;;

Feature		Fibrous protein	Globular protein
1	Level of organization	Secondary structure important; e.g. collagen	Tertiary structure important Quaternary structure might be present; e.g. enzyme
2	Molecular structure	Polypeptide chains are cross-linked at intervals to form long fibres or sheets;	Polypeptide chain tightly folded to form spherical shape;
3	Bonds	Mainly hydrogen bonds;	Interaction of the four types of bonds: Ionic bonds, hydrogen bonds, disulphide linkages and hydrophobic interactions;
4	Regularity of amino acid sequence	Quite regular e.g. collagen: Every third amino acid is glycine;	Rarely exhibit regularities in amino acid sequence;
5	Length of polypeptide chain	May vary in two samples of the same fibrous protein;	Is always identical in two samples of the same globular protein;
6	Solubility	Inability to form hydrogen bonds with water, insoluble in water;	Able to form hydrogen bonds with water, soluble in water;
7	Role in organism	Structural and support functions: Keratin, collagen;	Metabolic functions: Enzymes, antibodies, hormones e.g. insulin;

(b) Explain what is meant by primary, secondary, tertiary and quaternary structure of haemoglobin. [8]

1. The primary structure of a protein molecule refers to its number, type and sequence of amino acids held together by peptide bonds in the linear strand of polypeptide chain;;
2. Haemoglobin consists of 4 polypeptide chains: 2 α chains and 2 β chains;;
3. Each α chain has a specific sequence of 141 amino acids while each β chain has a specific sequence of 146 amino acids;;
4. Secondary structures are geometrically regular repeating structures, including α -helix and the β -pleated sheet, stabilised by hydrogen bonds between groups in the main chain of the polypeptide;;
5. Tertiary structure is the three-dimensional conformation of a polypeptide maintained by ionic bonds, hydrogen bonds, disulphide bonds and hydrophobic interactions;;
6. Each polypeptide chain of haemoglobin coils into an α -helix;;
7. The α -helix folded upon itself into a spherical globular protein maintained by interactions between the R-groups of its amino acids. The shape of the haemoglobin is held by tertiary bonds - hydrogen bonds, ionic bonds, disulphide bonds and hydrophobic interactions;;
8. Quaternary structure consists of an aggregation of 2 or more polypeptide chains held together by hydrophobic interactions, disulphide linkages, hydrogen bonds and ionic bonds;;
9. Haemoglobin's four polypeptide chains pack closely together, held together by hydrophobic interactions, disulphide linkages, hydrogen bonds and ionic bonds resulting in a nearly spherical haemoglobin molecule;;

Pt 8-9: bonds need to be mentioned only once.

(c) With reference to DNA polymerase, describe its mode of action. [6]

1. DNA polymerase is a biological catalyst which speed up biochemical reactions by lowering the activation energy of the reaction;;
2. remains chemically unchanged at the end of the reactions/DNA replication they catalysed and can be reused;;
3. DNA polymerase reduces the activation energy by holding substrates close together at the correct angle and orientation for successful interaction and collision OR straining the chemical bonds within the substrates until they break;;
4. DNA polymerase is highly specific in the reactions catalysed because only substrates of a complementary shape will fit into the active site with distinctive configuration –spatial fit;;
5. DNA polymerase is only able to add nucleotides to an 3'-OH group of another nucleotide due to the specificity of its active site;;
6. Resulting in the formation of phosphodiester bonds between nucleotides of DNA strand;;
7. Other relevant points- induced fit hypothesis, lock and key hypothesis

- 9 (a) Describe the transmission of an action potential along a myelinated neurone. [8]

A. Resting Membrane Potential ("R" – 1)

1. At resting potential (-70mV), there is a high concentration of Na⁺ ions outside and a high concentration of K⁺ ions inside the neurones. This is maintained by the sodium potassium pump which pumps 3 Na⁺ ions out of axon and 2 K⁺ ions into the axon via active transport using energy in the form of ATP;;

B. Depolarisation/Action Potential ("D" – max 4)

2. When the neurone is stimulated, voltage-gated Na⁺ channels in the node of Ranvier open, causing an influx of Na⁺ ions into the axon down a concentration gradient;;
3. This causes depolarisation of the membrane to threshold potential (-50mV);;
4. Once threshold potential is reached, additional voltage-gated Na⁺ channels open; more Na⁺ ions enter the axon and membrane is depolarised to +40mV, generating an action potential at that site;;
5. Some of the local current generated by the action potential will then flow passively down the axon to the next node of Ranvier, setting up a local circuit;;
6. This passive current flow depolarizes the membrane in the next node of Ranvier of the axon, thus opening the voltage-gated Na⁺ channels in that node of membrane and triggering an action potential at this node;;
7. This process repeats until the end of the axon;;

Points 2, 3, 4, 5, 6, needed to gain full credit for this section

C. Saltatory Conduction ("S" – max 1)

8. Depolarisation is restricted only to the nodes of Ranvier in myelinated axons. The voltage-gated ion channels that produce the action potential are concentrated (at high density) in the nodes of Ranvier. Also, extracellular fluid is in contact with the axon membrane only at the nodes, so that the flow of ions between the inside and outside of the axon can occur only in these regions;;
9. Thus action potential "jumps" from one node of Ranvier to another down the axon. This mechanism is called saltatory conduction;;

D. Repolarisation ("RE" – max 1)

10. Behind the impulse, the voltage-gated Na^+ channels close and the voltage-gated K^+ channels open, K^+ ions begin to diffuse out of the axon along a concentration gradient;;
11. The outflux of K^+ causes the membrane behind the impulse to become repolarised before resting potential is restored;;
12. After the impulse has passed and the neurone is repolarised, Na^+ ions are once again actively expelled by the Na^+-K^+ pump to increase the external concentration of Na^+ and so allow the passage of another impulse;;

E. Hyperpolarisation/Refractory Period ("H" – max 1)

13. Voltage-gated K^+ channels remain open for a while longer after the repolarisation phase;;
14. There is thus a slight 'overshoot' in the movement of K^+ ions which causes the membrane potential to become slightly more negative than its normal resting potential, this is called hyperpolarisation;;
15. Within another millisecond or two, the resting state is restored, and the system is ready to respond to another stimulus;;

(b) Explain how a nerve impulse is passed across a synapse.

[8]

A. Depolarisation of pre-synaptic membrane ("D1" – 2)

1. Before the presynaptic membrane is depolarised, there are few Ca^{2+} ions inside the synaptic knob;;
2. The arrival of an impulse/action potential at the synaptic knob causes voltage-gated Na^+ channels and voltage-gated Ca^{2+} channels to open, resulting in influx of Na^+ ions and Ca^{2+} ions into the synaptic knob from the synaptic cleft;;

B. Release of acetylcholine ("A" – max 2)

3. The influx of Ca^{2+} ions causes vesicles containing neurotransmitter—acetylcholine to move to the presynaptic membrane and fuse with it, emptying the neurotransmitters into the synaptic cleft by exocytosis;;
4. Following this, the Ca^{2+} ions are pumped out of the synaptic knob by active transport using ATP;;
5. Acetylcholine diffuse across the 20nm synaptic cleft, usually in less than 0.5 ms;;

C. Depolarisation of post-synaptic membrane ("D2" – max 4)

6. Acetylcholine is complementary in shape to specific binding sites of receptors on the postsynaptic membrane, and recognises and binds to the binding sites;;
7. This changes the shape of the receptor protein, opening chemically-gated Na^+ channels on the postsynaptic membrane;;
8. Influx of Na^+ ions into the cytoplasm along concentration gradient depolarise the postsynaptic membrane;;
9. The membrane potential becomes more positive than resting potential;;

10. If the membrane potential of the postsynaptic membrane reaches threshold potential (-50mV), an action potential is generated in the postsynaptic neurone and travels down the axon to the next synapse or to an end plate;;

11. The local potential that develops is known as an excitatory postsynaptic potential (EPSP).

(c) Suggest the reasons why nerve impulses only travel in one direction across the synapse.
[4]

1. Ca^{2+} ion channels present only in the pre-synaptic knob;;
 2. Vesicles containing neurotransmitters are present only in the pre-synaptic knob;;
 3. Neurotransmitters can only be released from the pre-synaptic knob and they will diffuse across the synaptic cleft to the post-synaptic membrane;;
 4. Receptors for the neurotransmitters are only found on the post-synaptic membrane;;
 5. Therefore only the post-synaptic membrane will be depolarised upon the binding of the neurotransmitters;;
- Impulse can therefore only travel in one direction across the synapse.

R: references related to one directional transmission of nerve impulses along axons

Max = 4